Siddharth P.

AI Systems Engineer | Agentic AI | Enterprise AI Infrastructure | AI Product Engineering

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PROFILE

AI Systems Engineer focused on agentic applications, enterprise AI workflows, and product engineering. Experience and project work span tool-connected systems, human review boundaries, local retrieval, applied ML research, and user-facing web/mobile applications. Project status and attribution are stated at the level supported by available records.

EXPERIENCE

FDE Intern - Subhanu | Current

- Working across the Enterprise AI platform (LibreChat and its agents), the KiddyPi app, and a Voice Agent.

AI/ML Intern - Equvinoxis Pvt Ltd | Jun 2025 - Jul 2025

- Researched non-invasive facial-analysis approaches for stress and fatigue monitoring in pilots under prolonged aviation-stress conditions, within a human-lab research setting.
- Translated research objectives into practical monitoring workflows and contributed to the applied-research direction for human-state monitoring.

EDUCATION

B.E. Artificial Intelligence & Machine Learning - B.N.M. Institute of Technology, Bengaluru | 2023-2027 | CGPA 8.61/10

Pre-University (PCMC), Sri Chaitanya PU College | 2021-2023 | ICSE, Sri Vani Education Centre | 2021

TECHNICAL CAPABILITIES

Agent & tool systems: MCP, LangGraph, tool workflows, human review, audit evidence | **AI & retrieval:** local LLMs, Ollama, RAG, BM25, Chroma, citations, abstention

Backend & data: Python, FastAPI, Flask, TypeScript, React, Next.js, Node.js, SQL, SQLite, PostgreSQL, Firebase, Supabase | **Product & edge:** Flutter, Dart, Kotlin, Android, MediaPipe, Raspberry Pi

Delivery & evaluation: Docker, Git, GitHub Actions, CI, pytest, model evaluation, reproducible experiments, privacy-aware and local-first design

ENGINEERING THEMES

Evidence before certainty; explicit human decision boundaries; local execution where control matters; reproducible evaluation with limitations documented.

SELECTED TECHNICAL WORK

CREDA | Shipped product

Skill-verification product publicly described around project-based interviews, repository analysis, and shareable skill records. Individual feature ownership is not established by the public product page. Technologies: Public product page; implementation stack not asserted. [proof]

TrustLayer | Shipped product

Paste-and-analyze experience that surfaces sentence-level confidence signals and explains its visible, rule-based decision path. The project describes itself as client-side and without API calls; this is not a truth-verification guarantee. Technologies: React, Tailwind CSS, Vite. [proof]

AARNA | Team-built RIFT 2026 testnet

Decentralized blue-carbon MRV workflow modeling project submission, IPFS evidence, validator review, Algorand testnet credit issuance, registry, and marketplace. Team-built; public artifacts do not partition individual contribution. No claim of real-world credit issuance. Technologies: Algorand, AlgoPy, Puya, IPFS, React, TypeScript. [proof]

SmartHelm | Android companion app exists; release unverified

Edge-oriented rider-safety system. Public source includes a Kotlin Android application alongside MediaPipe eye-state processing, a Raspberry Pi path, optional MAX30102 inputs, and Firebase-connected fleet components. Release status is not verified. Technologies: Kotlin, Android, MediaPipe, Python/Flask, Raspberry Pi, Firebase. [proof]

SELECTED TECHNICAL WORK | CONTINUED

Prototypes, active builds, product work, and capstone evidence

PayGuard-AgentX | Prototype
Local-first retail operations prototype for restock, invoice, and fraud-investigation workflows. Project source describes supervised agent routing, MCP tool contracts, evidence records, and human review; it is not represented as a production deployment. Technologies: Python, LangGraph, MCP, SQLite, Chroma, Kuzu, Ollama. [proof]

OriginX-T (ORIGIN-T) | Active research capstone
Dataset-forensics and early-warning research for LLM fine-tuning risk. Implemented core includes controlled contamination experiments, LoRA runs/measurements, and calibrated risk classification; advanced survival, conformal, and white-box layers remain gated. Technologies: Python, PyTorch, Transformers, PEFT, scikit-learn, MLflow. [proof]

NirmiqCodeSensei | Active local-first build
Repository learning tool with eight-lens code review, architecture mapping, code-grounded DSA, explain-back practice, local storage, and MCP connections. README describes localhost-only use and no telemetry. Technologies: TypeScript, Next.js, SQLite, Drizzle, MCP. [proof]

NirmiqResearchOS | Active local RAG MVP
Offline-first academic workspace with local retrieval, citations, Paper Lab, and Exam Lab. Repository documents a demo/MVP and local generation path, not a hosted multi-user SaaS. Technologies: Python, FastAPI, Next.js, SQLite, BM25, Chroma, Ollama. [proof]

AscensionOS | Product build
Mobile-first personal dashboard with local cache and offline queue, optional synchronization, PWA behavior, and Android packaging work. Technologies: Next.js, TypeScript, Tailwind, Supabase, IndexedDB, Capacitor. [proof]

Vibecode Project Manager | Public developer workflow tool
Approval-first skill routing that analyzes a project and selects relevant capabilities; source includes deterministic ranking and cross-platform shell support. Technologies: Bash, PowerShell, GitHub Actions. [proof]

ACADEMIC WORK

Multimodal stress/fatigue review | BNMIT academic review
Supplied CV records literature survey, requirement specifications, architecture design, and conclusion contributions to an IEEE-format review paper. Technologies: Literature review and system architecture.

ADDITIONAL VERIFIED SYSTEMS

NirmiqEcho (Windows) | Local-first assistant
Offline voice assistant with separate command and dictation modes; project notes document faster-whisper and local Ollama fallback. Technologies: Python, faster-whisper, Ollama. [proof]

NirmiqEcho-Mobile | Android MVP
Separate Android speech companion using system speech recognition with an offline-preference path; distinct from the Windows assistant. Technologies: Kotlin, Android.

VyomaGuard PHM | Mission-assurance prototype
Explainable space-mission triage prototype combining telemetry health signals and conjunction-risk ranking; explicitly not flight-certified. Technologies: Python, scikit-learn, Isolation Forest, SGP4/TLE, Pydantic, Streamlit. [proof]

AGION | Flutter product build
Personal productivity and goal-tracking application described in the supplied CV and local project README. Technologies: Flutter, Dart, Riverpod, Hive, Firebase. [proof]

PROJECT INDEX & RELATED WORK

Aliases are merged into one entry. Public repository presence is proof of an artifact, not by itself proof of individual ownership or release.

ASSESSMENT

FeeOps Finance Agent | Subhanu technical assessment; separate from employment
Scoped assessment prototype. It is not presented as Subhanu employment output; external technical detail is omitted. Technologies: Assessment evidence kept private.

PUBLIC SYSTEMS AND CODE PROJECTS

vibecode-orchestrator | Public repository / developer tooling
Intent-based skill routing and per-lane arbitration for large assistant skill libraries; summarized in the supplied CV. Technologies: Skill routing; implementation stack not independently enumerated. [proof]

Stress-Detection-System | Computer-vision study
Webcam-based stress/fatigue work using facial, pulse, HRV, PERCLOS, and blink signals as summarized in the supplied CV. Technologies: Python, computer vision, machine learning. [proof]

Cognitive stress detector | Separate model study
Distinct public repository for multimodel cognitive-stress/fatigue classification; detailed ownership and comparative metrics are not asserted. Technologies: Deep-learning model evaluation. [proof]

Automated Brain Tumour Detection | Research classifier
MRI classification study with a hybrid CNN/ViT and radiomics direction; performance figures from the old CV are intentionally omitted pending a reproducible evaluation audit. Technologies: PyTorch, ResNet50, Vision Transformer, radiomics, Grad-CAM. [proof]

AnonThera | Public application project
Student mental-health support concept/application with anonymous support and wellbeing workflows; maturity and personal feature ownership are not independently verified. Technologies: Public repository; stack not asserted. [proof]

PayGuard-DQ | Data-quality project
Payment/transaction data-quality work for identifying inconsistent or corrupt records; maturity and detailed implementation are not independently verified. Technologies: Public repository; stack not asserted. [proof]

NIDS | Network-security ML project
Machine-learning network intrusion-detection repository; claims are limited to the supplied CV and public project record. Technologies: Machine learning; stack not asserted. [proof]

Skylark BI Agent | Assignment/demo
Conversational analytics over monday.com boards with deterministic server-side metrics and an LLM for open-ended follow-up; no board identifiers or credentials included. Technologies: Next.js, TypeScript, GraphQL, Gemini. [proof]

Skylark GCP inference pipeline | Localization assignment
Manifest-driven, bounded-memory inference over aerial GeoTIFF imagery with schema and coordinate checks; assignment scope, not a full-scene production claim. Technologies: Python, ONNX Runtime, Docker, pytest. [proof]

PostPilot AI | Coursework / PE3
Social engagement forecasting pipeline; the local project record notes synthetic-data limitations and weak results, so no accuracy claim is made. Technologies: Python, machine learning, social-data analysis. [proof]

SocialGuard-R | Coursework / PE4
R/Shiny dashboard for exploring unusual social-media engagement patterns; README notes that anomaly scores are not proof of fraud. Technologies: R, Shiny, bslib, dplyr, ggplot2.

HACKATHON AND RESEARCH

BOLT-Gemma B-lite | Kaggle/DeepMind paper-track study
Paper-only exploratory study of context-capsule/reveal decisions under token budgets. The local record says the model was not fine-tuned and held-out/model feasibility remain unverified. Technologies: Research study; not a trained model release.

Enveda CASMI 2026 | Kaggle hackathon research / WIP
Research/code work remains gated by a free-memory requirement in the local record; no submission, winning, or production-admission claim is made. Technologies: Computational chemistry / ML; exact stack not asserted.